Conservation Genetics Practice Problems #2

1) List three factors that can cause linkage disequilibrium.

2) Assume a population is in Hardy-Weinberg Equilibrium at two biallelic loci—A and B—where alleles A_1 and B_1 are on the same chromosome and perfectly associated at time $t = 0$, and alleles A_2 and B_2 are on the same chromosome and perfectly associated at time $t = 0$. After 3 generations of meiosis with a recombination rate of $c = 0.2$, which of the following statements is true?

   a) allele frequencies have changed but haplotype frequencies have not; evolution has occurred
   b) allele frequencies have changed but haplotype frequencies have not; evolution has not occurred
   c) haplotype frequencies have changed but allele frequencies have not; evolution has occurred
   d) haplotype frequencies have changed but allele frequencies have not; evolution has not occurred

3) A sample ($n = 20$) from a population of small wriggly critters is genotyped at biallelic loci A and B, producing the following haplotype counts: A_1B_1=5, A_1B_2=2, A_2B_1=3, and A_2B_2=5. Using these data, calculate D.

4) Why is the "equilibrium" in linkage disequilibrium different than the "equilibrium" in Hardy-Weinberg Equilibrium?

5) A sample ($n = 10$) of a population of large sedate critters is genotyped at biallelic loci A and B, producing the following individual genotypes: A_1A_1/B_1B_1 (3), A_1A_2/B_1B_2 (3), A_2A_2/B_1B_1 (1), A_1A_2/B_2B_1 (3). Evaluate the following statements:

   a) This population is in Hardy-Weinberg Equilibrium at locus A (**T**/**F**)
   b) This population is in Hardy-Weinberg Equilibrium at locus B (**T**/**F**)
   c) Alleles at locus A and B assort independently (**T**/**F**)